

PROJECT DOCUMENTATION

Hospital Operations and Patient Analytics Dashboard

Module 8 Project Documentation Report

Project: Hospital Operations and Patient Analytics Dashboard – Group 2

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Technology: Tableau Desktop / Tableau Public, Python, Pandas, NumPy, Tableau Hyper

GitHub Branch: Subashree-Jeyaraman

Academic Project Submission • Module 8

Document Control

Item	Details
Project Title	Hospital Operations and Patient Analytics Dashboard
Project	Group 2
Prepared By	Subashree Jeyaraman
Module	Module 8 – Documentation and Project Delivery
Primary Specification	HOPA.pdf – Module 8 requirements document
Dashboard Artefacts	Four supplied dashboard captures
Final Workbook Specified by Requirements	MedTrack_DV.twbx / single Tableau workbook (.twbx)
GitHub Branch	Subashree-Jeyaraman
GitHub Repository	https://github.com/springboardmentorpl/Hospital-Operations-and-Patient-Analytics-Dashboard-Group-2/tree/Subashree-Jeyaraman

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1. Project Overview

Project purpose

The Hospital Operations and Patient Analytics Dashboard is a Tableau-based healthcare operations analytics project designed to transform hospital operational and patient admission data into interactive views for monitoring hospital performance, patient flow, department efficiency, and healthcare resource utilisation. The supplied Module 8 requirements describe a unified Tableau workbook containing four interconnected dashboard areas and position the solution for use by hospital administrators, healthcare managers, medical staff, and policymakers.

Project scope

The project scope covers healthcare data collection and integration, data cleaning and transformation, healthcare KPI engineering, dashboard planning, dashboard development, dashboard integration, testing and validation, and final documentation and delivery. The analytical purpose is operational decision support rather than clinical diagnosis or treatment.

Source basis and artefact note

This report uses the supplied Module 8 requirements document (HOPA.pdf) as the primary specification and the supplied dashboard captures as visual evidence of the implemented dashboard suite. The requirements document identifies the project as “MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)” and specifies the final deliverable as a single Tableau workbook (.twbx). Underlying Tableau workbook metadata, including the exact internal worksheet names and calculated-field formulas, is not explicitly exposed in the supplied artefacts available for documentation; where such detail cannot be verified, this report states so rather than inferring it.

2. Dataset Sources

Source categories

The project requirements identify publicly available Hospital Datasets and Patient Admission Datasets as the source basis. The data-collection scope specifically includes hospital operational datasets, patient admission records, and department and resource data, followed by integration of the healthcare datasets.

Hospital operational data

Hospital operational data provides the operational context required for hospital-level monitoring, including measures associated with admissions, occupancy, length of stay, and resource/capacity analysis. The exact originating dataset name and source URL are not explicitly available in the supplied project requirements.

Patient admission data

Patient admission records support patient-flow analysis, admission trends, discharge tracking, average-stay analysis, demographic analysis, admission-type analysis, and diagnosis-related views where the corresponding fields are present in the dashboard data.

Department data

Department data supports department-level patient volume, performance comparison, efficiency analysis, revenue analysis, treatment-capacity views, and department/resource relationships. The requirements explicitly call for gathering department data and developing department analytics.

Resource data

Resource data supports bed utilisation, staff allocation monitoring, equipment utilisation tracking, resource availability analysis, and capacity-planning insights. The Resource Utilisation dashboard capture also shows resource distribution and ward-type utilisation views.

Data integration

The requirements specify integration of healthcare datasets before transformation into Tableau-ready data. The resulting analytical layer is intended to provide a common basis for hospital, patient, department, and resource views within the unified workbook.

Data model and important fields

Fields visible in the supplied dashboard captures include department name, admission date, discharge date, admission type, ward type, gender, age group, diagnosis, resource type, and measures associated with beds, resources, staff, revenue, admissions, length of stay, occupancy, readmission, and efficiency. The exact physical schema, relationships, joins, and field-level data types are not explicitly available in the supplied requirements document.

Tableau-ready dataset / extract

The project requirements explicitly call for Tableau-ready datasets and identify `hospital_cleaned.csv` and `hospital_final_dataset.xlsx` as planned deliverables at earlier project stages. The final Tableau workbook is specified as the consolidated delivery artefact. A separately supplied final extract file is not available in the current artefact set, so its exact structure cannot be independently verified here.

Source limitations

Exact public dataset URLs, source publishers, extraction dates, row counts, and source-level provenance details are not explicitly available in the supplied project requirements. Accordingly, no unverified source names, URLs, record counts, or completeness statistics are introduced in this report.

3. Data Cleaning and Transformation

Cleaning objectives

The data-preparation stage is intended to produce consistent, complete, Tableau-ready healthcare operational data. The requirements explicitly include duplicate removal, missing patient-data handling, department-name standardisation, healthcare-indicator normalisation, and creation of Tableau-ready datasets.

Duplicate handling

Duplicate records are identified as a defined cleaning task and are to be removed before analytical modelling. The supplied requirements do not provide the number or percentage of duplicates removed.

Missing-value handling

Missing patient data is identified as a required cleaning task. The Module 1/2 evaluation target specifies less than 2% missing values after cleaning, but the supplied artefacts do not provide an achieved measured percentage; therefore no achievement percentage is claimed.

Department-name standardisation

Department names are standardised so that department-level aggregation and comparison use consistent categories. This supports the department performance and resource comparison views.

Healthcare-indicator normalisation

Healthcare indicators are normalised as part of the cleaning stage to support consistent operational metrics across the integrated data. The exact normalisation formulae are not explicitly available in the requirements document.

Date conversion and standardisation

Admission and discharge dates are used in the dashboard captures for monthly trends, discharge-year analysis, and date filtering. Date conversion/standardisation is therefore part of the analytical preparation evidenced by the dashboard design; the exact source-to-target conversion expression is not explicitly available.

Patient-related transformations

The dashboard suite uses patient-related analytical dimensions including age group, gender, admission type, diagnosis, admission date, discharge date, and length of stay. These represent transformed or analytical uses of patient attributes; the exact field-by-field transformation logic is not explicitly documented in the supplied requirements.

Length-of-stay calculation

Average Length of Stay is a required KPI, and the dashboard captures include average-stay views and age-versus-length-of-stay analysis. The precise calculation expression for individual stay duration is not explicitly provided in the supplied requirements; therefore the underlying formula is recorded here as not explicitly available rather than inferred.

Age grouping

Age groups are visibly used in the Patient Dashboard and include categories such as 0–17, 18–30, 31–45, 46–60, 61–75, and 76+. The supplied dashboard capture provides evidence of these analytical categories; the exact calculated-field expression used to assign a patient to a group is not explicitly available.

Stay categorisation

Length-of-stay analysis is present in the dashboard suite. A separate named stay-category field and its exact thresholds are not explicitly available in the supplied project requirements.

Billing categorisation

Revenue/billing analysis is visible in the Hospital and Department dashboard captures. A separate billing-category field and its exact classification rules are not explicitly available in the supplied artefacts.

Readmission flag

Readmission analysis is a stated project requirement and a KPI requirement. The exact source field or calculated logic used to identify a readmission is not explicitly available in the supplied requirements document.

Occupancy flag / occupancy metric

Occupancy monitoring and bed utilisation are explicit project requirements, and the dashboard captures show Bed Occupancy and Monthly Bed Occupancy Rate. The exact occupancy-flag logic and denominator used by the final workbook are not explicitly available in the supplied requirements.

Department efficiency score

Department Efficiency Score is explicitly identified as a KPI in the requirements and Department Efficiency is represented in the dashboard suite. The exact scoring formula and weighting are not explicitly available in the supplied artefacts.

Cleaned vs derived fields

Source fields are represented by directly collected attributes such as admission date, discharge date, department, gender, diagnosis, admission type, ward type, and resource type. Cleaned fields are standardised/normalised versions of source attributes. Derived analytical fields include KPI and grouping concepts such as length of stay, age group, readmission, occupancy, and department efficiency. Exact calculated-field formulas are not explicitly available in the supplied requirements.

4. KPI Definitions

KPI documentation note

The requirements specify KPI engineering and identify Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate, and Department Efficiency Score as required KPI calculations. The supplied dashboard captures additionally show dashboard-level measures such as total patients, total departments, total wards, total revenue, average patient age, average stay, occupancy, readmissions, and department efficiency. The table below documents the supported business meaning without fabricating result values.

KPI	Definition	Calculation / Logic	Business Purpose
Total Admissions	Number of admission events represented in the analytical data.	Admissions count/aggregation used by the dashboard; exact underlying Tableau aggregation is not explicitly available.	Monitor patient inflow and hospital workload.
Total Patients	Number of patients represented in the patient analytical population.	Patient count/aggregation represented by the dashboard; exact underlying aggregation is not explicitly available.	Understand overall patient volume.
Total Departments	Number of departments represented in the dashboard data.	Count of represented department categories.	Establish organisational scope for comparison.
Total Wards	Number of wards represented in the dashboard data.	Count of represented ward categories; exact field logic is not explicitly available.	Understand inpatient capacity structure.
Total Revenue	Total revenue represented in the analytical data.	Revenue aggregation shown in revenue-by-department views; exact source-field formula is not explicitly available.	Monitor financial contribution across departments.
Average Length of Stay	Average duration of patient stay represented in the dataset.	Average stay measure; exact individual stay formula is not explicitly available.	Monitor patient throughput and bed demand.
Average Patient Age	Average age of patients represented in the patient population.	Average of patient age measure; exact field-level implementation is not explicitly available.	Understand patient demographic profile.
Occupancy Rate / Bed Occupancy	Measure of occupied bed capacity relative to available capacity.	Occupancy calculation is required by the project; exact numerator/denominator is not explicitly available.	Monitor bed utilisation and capacity pressure.
Readmission Rate / Readmission Score	Measure representing readmission performance.	Readmission rate is a required KPI; exact readmission identification logic is not explicitly available.	Identify repeat-admission patterns for operational review.
Bed Utilization Rate	Measure of bed capacity utilisation.	Required KPI; exact formula is not explicitly available.	Support bed allocation and capacity planning.
Average Department Efficiency / Department Efficiency Score	Comparative measure of department operational efficiency.	Department Efficiency Score is a required KPI; exact scoring formula/weights are not explicitly available.	Compare department performance and identify operational differences.
Bed Days	Total patient-bed usage over the measured stay period.	Derived from patient stay duration and/or bed occupancy data; exact workbook formula is not explicitly available.	Support capacity and utilisation planning.

5. Dashboard Guide

Dashboard suite

The supplied Module 8 specification defines four dashboard areas: Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization. The supplied dashboard captures present these areas using the visible titles Hospital Dashboard, Patient Dashboard, Department Dashboard, and Resource Utilisation Dashboard. The four views collectively cover hospital performance, patient analytics, department performance, and resource utilisation.

Hospital Dashboard

Purpose: Provides a hospital-level operational overview. Key visuals visible in the supplied capture include revenue by department, monthly admissions trend, department versus admission type, department staff/head, age versus length of stay, and KPI cards for admissions, revenue, average stay, and bed occupancy. **Filters shown** include admission month, admission type, and department name. **Business use:** supports high-level monitoring of admissions, revenue, occupancy, staffing and department-level operational patterns.

Patient Dashboard

Purpose: Provides patient demographic and clinical-insight views. Key visuals include patients by age group, discharge year by gender, top diagnoses, patients by admission type, and admission-type distribution. **Filters shown** include gender and age group, with a visited-month filter at the top. **Business use:** supports understanding of patient composition, admission patterns, demographic distribution, diagnoses represented in the dataset, and patient-flow characteristics.

Department Dashboard

Purpose: Provides department performance and efficiency analysis. Key visuals include admissions by department, revenue by department, daily admissions by department, average stay by department, and department efficiency. A department filter is visible, and the top control shows admission type. **Business use:** enables comparison of patient volume, financial contribution, length of stay and efficiency across departments, supporting operational review and resource planning.

Resource Utilisation Dashboard

Purpose: Provides hospital capacity and resource utilisation analysis. Key visuals include bed capacity by department, resource type distribution, admissions by ward type, monthly bed occupancy rate, resources by department, and utilisation by ward type. A department filter and resource-type selector are visible. **Business use:** supports capacity planning, bed allocation, resource availability review, ward utilisation monitoring, and department/resource comparison.

Filters and interactions

The requirements explicitly call for filters, navigation, dashboard actions, department comparisons, global filters, navigation controls, parameter actions, and dashboard linking. The supplied captures visibly demonstrate dashboard navigation and contextual filters such as department, admission type, date/month, gender, age group, and resource type. Exact action names and parameter definitions are not explicitly available in the requirements artefact.

6. Dashboard Content Inventory

Inventory note

The inventory below uses visible dashboard/visual titles from the supplied captures and requirement terminology. Exact underlying Tableau worksheet names are not explicitly available in the supplied project artefacts available for documentation.

Dashboard	Visualization / Worksheet (visible title)	Purpose
Hospital Dashboard	Revenue by Department	Compares revenue contribution across departments.
Hospital Dashboard	Monthly Admissions Trend	Shows admissions over time for operational trend monitoring.
Hospital Dashboard	Department vs Admission Type	Compares departments across admission pathways.
Hospital Dashboard	Department Staff & Head	Shows department-level staff/headcount view.
Hospital Dashboard	Age vs Length of Stay	Relates age-group categories to stay duration patterns.
Hospital Dashboard	Admissions / Revenue / Avg Stay / Bed Occupancy KPI cards	Provides high-level hospital operational indicators.
Patient Dashboard	Patients by Age Group	Shows patient distribution across age categories.
Patient Dashboard	Discharge year by Gender	Shows discharge volumes across gender categories and years.
Patient Dashboard	Top Diagnoses	Ranks diagnoses represented in the patient dataset.
Patient Dashboard	Patients by Admission Type	Shows patient volume by admission pathway.

Dashboard	Visualization / Worksheet (visible title)	Purpose
Patient Dashboard	Admission Type Distribution	Shows the distribution of admission-related categories in the patient view.
Department Dashboard	Admissions by Department	Compares patient admission volume by department.
Department Dashboard	Revenue by Department	Compares department revenue.
Department Dashboard	Daily Admissions by Department	Shows daily admission volume by department.
Department Dashboard	Average Stay by Department	Compares average stay across departments.
Department Dashboard	Department Efficiency	Compares department efficiency.
Resource Utilisation Dashboard	Bed Capacity by Department	Compares bed capacity across departments.
Resource Utilisation Dashboard	Resource Type Distribution	Shows the distribution of resource types.
Resource Utilisation Dashboard	Admissions by Ward Type	Shows admissions across ward categories.
Resource Utilisation Dashboard	Monthly Bed Occupancy Rate	Tracks occupancy rate over time.
Resource Utilisation Dashboard	Resources by Department	Shows resource allocation/distribution across departments.
Resource Utilisation Dashboard	Utilisation by Ward Type	Compares utilisation across ward types.

7. Healthcare Operations Analytics Methodology

End-to-end workflow

The methodology follows the project workflow specified in the requirements: Data Collection → Data Cleaning & Transformation → Healthcare KPI Engineering → Dashboard Planning → Dashboard Development → Dashboard Integration → Testing & Validation → Documentation & Delivery.

Data Collection

Hospital operational datasets, patient admission records, and department/resource data are collected and integrated. The objective is to establish the data foundation required for operational analytics.

Data Cleaning & Transformation

The integrated data is cleaned through duplicate removal, missing-data handling, department standardisation, healthcare-indicator normalisation, and preparation of Tableau-ready datasets.

Healthcare KPI Engineering

Operational KPIs are engineered for admissions, occupancy, average stay, readmissions, bed utilisation, and department efficiency. These metrics convert raw operational data into measures suitable for monitoring and comparison.

Dashboard Planning

Dashboard areas are planned around hospital overview, patient flow, department analytics, and resource utilisation. Requirements include filters, navigation, dashboard actions, and department comparisons.

Dashboard Development

The analytical views are implemented as interactive Tableau dashboards. The supplied captures evidence the four dashboard areas and their operational visualisations.

Dashboard Integration

The requirements specify global filters, navigation controls, parameter actions, and dashboard linking so that users can move between complementary analytical perspectives.

Testing & Validation

KPI calculations, healthcare metrics, dashboard interactions, and patient-flow analytics are to be validated before delivery. The requirements specify KPI accuracy above 95% as a Module 7 evaluation target; this report does not claim a measured achieved percentage because none is explicitly supplied.

Documentation & Delivery

The final stage packages the project into its documented submission structure, GitHub repository, and Tableau workbook. The requirements identify documentation and a portfolio-ready dashboard suite as the delivery outcome.

Operational analytics value

The methodology supports patient-flow monitoring through admissions, discharge, and stay analysis; hospital performance monitoring through KPI and revenue views; department efficiency through comparative operational metrics; resource utilisation through bed, staff, equipment and capacity views; and capacity planning through resource availability and utilisation analysis. The system is an operational analytics solution and is not presented as a clinical diagnosis or treatment system.

8. Dashboard Integration

Global filters

Global filters are explicitly required by the project specification. They provide a consistent way to constrain the analytical context across integrated dashboards where configured.

Department filters

Department-level filtering is visibly present in the supplied dashboard captures and supports focused comparison of hospital functions, patient volume, revenue, efficiency and resources.

Admission-type filters

Admission type is used as a filtering dimension in the dashboard suite and is visible in the Hospital and Department dashboard captures. It supports analysis of different admission pathways.

Date and month filters

Date/month filtering is represented through controls such as Admission Date, Admission Month, Visited Month, and monthly trend views in the supplied captures. These controls support time-based operational analysis.

Navigation

The supplied captures show a common navigation structure across Hospital Dashboard, Patient Analytics, Department Performance, and Resource Utilisation. The requirements explicitly call for navigation controls and dashboard linking.

Dashboard actions and parameter actions

Parameter actions and dashboard actions are explicitly included in the integration requirements. The exact action definitions and parameter names are not explicitly available in the supplied artefacts.

Cross-dashboard analysis

The integrated suite allows users to move from hospital-level context to patient, department and resource perspectives. This structure reduces the need to interpret operational measures in isolation and supports cross-functional review.

9. Testing and Validation

QA approach

Testing is structured around the Module 7 requirements: KPI calculation validation, healthcare-metric verification, dashboard-interaction testing, and patient-flow validation. The final documentation records the expected validation approach without inventing measured pass rates.

Test Area	Validation Method	Expected Result
KPI calculation validation	Recalculate selected KPI logic independently and compare with dashboard outputs.	KPI outputs match the validated calculation logic.
Data consistency	Check department, date, patient, admission and resource dimensions for consistent categories and formats.	No material inconsistencies remain in analytical categories.
Filter testing	Apply department, admission type, date/month, gender, age-group and resource-type filters where available.	Relevant dashboard views update consistently with the selected context.
Navigation testing	Use dashboard navigation controls and linked dashboard paths.	Users can move between the integrated dashboard areas without broken navigation.
Dashboard interaction testing	Test configured dashboard actions and parameter-driven interactions where present.	Configured interactions respond correctly and do not produce unexpected states.
Patient-flow validation	Compare admissions, discharge, stay and patient-flow views for logical consistency.	Patient-flow views represent the underlying time and patient dimensions consistently.

Test Area	Validation Method	Expected Result
Department analysis validation	Cross-check department admissions, revenue, average stay and efficiency views.	Department comparisons remain aligned across relevant views.
Resource analysis validation	Cross-check bed capacity, occupancy, resource distribution and ward utilisation views.	Resource and capacity views are internally consistent.
Workbook packaging validation	Open and inspect the final .twbx package and confirm embedded/supporting data dependencies.	Workbook opens successfully with required dashboard content and dependencies.
Module 7 accuracy criterion	Treat KPI accuracy above 95% as the stated evaluation requirement.	No achieved percentage is claimed unless supported by recorded QA evidence.

10. Project Structure

Repository structure

The Module 8 requirements specify the following project organisation:

```
/scripts
/data
/dashboard
/docs
```

The structure separates data-processing code, source/cleaned data, Tableau dashboard artefacts, and project documentation.

Scripts

The /scripts folder is intended for Python/data-processing utilities. The requirements identify `data_collection.py`, `hospital_cleaning.ipynb`, and `generate_hospital_kpis.py` as project-stage deliverables.

Data

The /data folder is intended for dataset artefacts. The requirements identify `hospital_raw_data.csv`, `hospital_cleaned.csv`, and `hospital_final_dataset.xlsx` as stage deliverables.

Dashboard

The /dashboard folder is intended for Tableau workbook/prototype artefacts. The requirements identify `medtrack_prototype.twbx`, `medtrack_dashboard_v1.twbx`, and `MedTrack_DV.twbx` across project milestones.

Docs

The /docs folder is intended for documentation and project-delivery materials, including the final Module 8 documentation and related QA/reporting artefacts where maintained.

GitHub repository

Project: Hospital Operations and Patient Analytics Dashboard – Group 2

Branch: Subashree-Jeyaraman

Repository: <https://github.com/springboardmentorpl/Hospital-Operations-and-Patient-Analytics-Dashboard-Group-2/tree/Subashree-Jeyaraman>

The supplied project specification requires GitHub repository deployment as part of Module 8 delivery.

11. Technical Stack

Technology overview

The technical stack below is restricted to technologies explicitly identified by the project requirements and the project information supplied for this documentation.

Area	Technology	Purpose
Data collection	Python, Hospital Dataset, Patient Admission Dataset	Collect and integrate healthcare source data.
Data processing	Pandas, NumPy	Process and transform healthcare datasets.
Data cleaning	Python	Perform cleaning and preparation tasks.
Visualization	Tableau Desktop / Tableau Public	Develop interactive dashboards and analytical views.

Area	Technology	Purpose
Dashboard integration	Tableau Filters, Parameters, Actions	Enable filtering, navigation, actions and integrated analysis.
Data extract / workbook	Tableau Hyper	Support Tableau-ready analytical data/extract workflows.
Version control / delivery	GitHub	Host and organise project code, data artefacts and documentation.
Documentation	Markdown	Support project documentation and repository-based reporting.

12. Final Deliverables

Delivery set

The project delivery is centred on a documented, portfolio-ready Tableau dashboard suite supported by cleaned/transformed data and KPI engineering artefacts. The requirements specify the GitHub repository, final documentation, and Tableau workbook as Module 8 deliverables.

13. Limitations

Operational limitations

The analytical outputs depend on the quality, completeness and consistency of the source data. The supplied requirements define data-cleaning targets but do not provide a measured final completeness percentage in the available artefacts.

Interpretation limitations

Dashboard metrics are intended for aggregate operational analysis and comparison. KPI interpretation should follow the field definitions and aggregation logic configured in the final workbook.

Derived KPI limitations

Several dashboard concepts—such as length of stay, readmission, occupancy and department efficiency—are represented in the project, but their exact calculated-field formulas are not explicitly documented in the supplied requirements. They should therefore be interpreted according to the final workbook configuration.

Use limitation

The project is an operational analytics dashboard. It is not a clinical diagnosis, treatment recommendation, or patient-care decision system.

14. Module 8 Compliance Checklist

Compliance summary

The documentation is structured directly around the Module 8 requirements. Status indicates documentation coverage, not an independent audit of the underlying workbook or repository contents.

Module 8 Requirement	Documentation Section	Status
Dataset Sources	Section 2 – Dataset Sources	Covered
KPI Definitions	Section 4 – KPI Definitions	Covered
Dashboard Guide	Section 5 – Dashboard Guide	Covered
Healthcare Operations Methodology	Section 7 – Healthcare Operations Analytics Methodology	Covered
Project Structure	Section 10 – Project Structure	Covered
GitHub Repository	Section 10 – Project Structure / GitHub repository	Covered
Final Documentation	This Module 8 report / Section 12 – Final Deliverables	Covered
Tableau Workbook	Section 12 – Final Deliverables; workbook is the specified final artefact	Covered
Portfolio Readiness	Section 14 – Module 8 Compliance Checklist; requirements identify a portfolio-ready dashboard suite	Covered

Source and Verification Note

This report intentionally avoids unverified dataset statistics, achieved cleaning percentages, KPI result values, calculated-field formulas, worksheet metadata, or source URLs. The supplied Module 8 requirements document provides the project specification and evaluation targets, while the supplied dashboard captures provide visual evidence of the dashboard areas and analytical views. Where an implementation detail cannot be verified from those artefacts, it is explicitly identified as not available in the supplied project artefacts.

Requirements evidence: the Module 8 specification defines dataset collection/integration, cleaning tasks, KPI engineering, four dashboard areas, integration controls, testing requirements, project structure, GitHub/Tableau Public delivery, and portfolio readiness. In particular, the specification requires documentation of dataset sources, KPI definitions, dashboard guide, and healthcare operations methodology, and identifies the final Tableau workbook as a delivery artefact. (HOPA.pdf, pp. 1–10.)